
Toscanini 03 buyer decision card

inv:r452-fresh-evaporative-vaccine-cooler:ae7a00faa34a · run run · maturity EXPERIMENT_READY · 3D class ENGINEERING_3D

What is it?

A recorded invention candidate (inv:r452-fresh-evaporative-vaccine-cooler:ae7a00faa34a) for the problem: • Problem Id: r452-fresh-evaporative-vaccine-cooler

- Device: passive evaporative-cooled vaccine storage vessel for last-mile clinics
- Failure: vaccine doses spoil in transit because the passive evaporative cooler loses cooling margin within hours when ambient humidity rises, the water-wick channel... — the proposed intervention: • Mechanism: A composite radiative coating embedded with microencapsulated phase-change materials that alter their infrared emissivity based on temperature thresholds
- Intervention: Apply a dual-layer coating where the base layer provides constant radiative cooling while the embedded phase-change l...

What is actually established?

The architecture survived the recorded adversarial challenge, adjudication and evolution stages; status EVOLVED_INVENTION_CANDIDATE. Evidence by class: direct support: 3; partial support: 5; background: 1; analogy: 4; contradictory: 1; irrelevant: 1. Model-derived engineering content is labeled MODEL_DERIVED; nothing claims physical validation (no PHYSICAL_OBSERVATION exists in this package).

What is not established?

FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005 — 7 recorded unknowns, each with a resolution action in UNKNOWN_ROADMAP.json. No numeric design decision is possible until the blockers are sourced.

What could kill it?

the falsification rule recorded in the decisive experiment contract

Why does it matter?

reliable operation of passive evaporative-cooled vaccine storage vessel for last-mile clinics

What is different?

- Collision Novelty Risk: UNRESOLVED_INSUFFICIENT_EVIDENCE
- Prior Art Status: UNRESOLVED_INSUFFICIENT_EVIDENCE
- Multi Source Directions: • Discovery: (sources_recorded, sources_errored)
- Destruction: (sources_recorded, sources_errored)
- Transfer: (sources_recorded, sources_errored)
- Reality: (s...

What would we build?

- WP-01: test article first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed); measurement physical bench/clinical experiment — NOT simulation
- WP-02: test article constraint-compliance envelope article; measurement constraint-compliance characterization

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- WP-03: test article article for VF-001; measurement Compare thermal performance of coated vs uncooled units under controlled temperature cycling from 15°C to 45°C ambient
 - WP-04: test article article for VF-002; measurement run killer experiment: falsification_test_from_candidate (EIG/cost 0.4557178723356231)
 - WP-05: test article article for VF-003; measurement temperature-rise mapping at max duty
 - WP-06: test article article for VF-004; measurement temperature-rise mapping at maximum duty with IR or embedded sensing (measured quantity: temperature rise)
 - WP-07: test article article for VF-005; measurement thermal-resistance trend over accelerated cycling (measured quantity: mechanical stress / fatigue endurance)
 - WP-08: test article article for VF-006; measurement temperature-rise mapping at max duty
 - WP-09: test article article for VF-007; measurement worst-case ambient soak with staged power steps and stability monitoring (measured quantity: power / energy deposition)

What must the buyer bring?

canonical problem context: UNKNOWN

buyer type: UNKNOWN — buyer class not determinable from the canonical problem text [UNKNOWN]

engineering capability: UNKNOWN — required engineering capability not determinable without a context decision [UNKNOWN]

manufacturing capability: UNKNOWN — required manufacturing capability not determinable without a context decision [UNKNOWN]

regulatory: UNKNOWN — regulatory applicability not determinable from the canonical problem text [UNKNOWN]

market channels: UNKNOWN — market channels not determinable without a context decision [UNKNOWN]

What is the cheapest decisive next experiment?

physical bench/clinical experiment — NOT simulation

What should we do next?

Fund and run the recorded decisive experiment (physical bench/clinical experiment — NOT simulation) — the pre-registered rule either kills the architecture or graduates it to a sourced measurement.

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.